In [1]:
```
# importing major libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [3]:
```
df = pd.read_csv('loan_approval_dataset.csv').drop(columns='loan_id')
```

In [4]:
```
df.head()
```

Out[4]:

	no_of_dependents	education	self_employed	income_annum	loan_amount	l
0	2	Graduate	No	9600000	29900000	
1	0	Not Graduate	Yes	4100000	12200000	
2	3	Graduate	No	9100000	29700000	
3	3	Graduate	No	8200000	30700000	
4	5	Not Graduate	Yes	9800000	24200000	

In [5]:
```
df.shape
```

Out[5]:
```
(4269, 12)
```

In [6]:
```
# df.loan_status
df.columns.str.strip()
```

Out[6]:
```
Index(['no_of_dependents', 'education', 'self_employed', 'income_annum',
       'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value',
       'commercial_assets_value', 'luxury_assets_value', 'bank_asset_value',
       'loan_status'],
      dtype='object')
```

In [7]:
```
df.columns = df.columns.str.strip()
```

In [8]:
```
df.columns
```

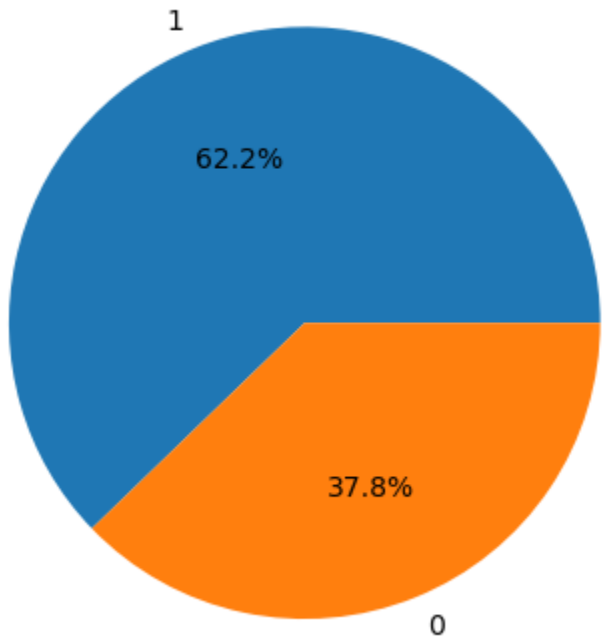
Out[8]:
```
Index(['no_of_dependents', 'education', 'self_employed', 'income_annum',
       'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value',
       'commercial_assets_value', 'luxury_assets_value', 'bank_asset_value',
       'loan_status'],
      dtype='object')
```

In [9]:
```
df.loan_status.unique()
```

Out[9]: array([' Approved', ' Rejected'], dtype=object)

In [10]:
```
df.loan_status = df.loan_status.map({' Approved':1, ' Rejected':0})
```

In [11]:
```
plt.pie(df.loan_status.value_counts(),labels=df.loan_status.value_counts().inc
plt.show()
df.loan_status.value_counts()
```



Out[11]:
```
loan_status
1    2656
0    1613
Name: count, dtype: int64
```

In [12]:
```
df.education.unique()

df.education = df.education.str.strip()
df.self_employed = df.self_employed.str.strip()
```

In [13]:
```
# column transformer
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
l1 = ['Graduate', 'Not Graduate']
l2 = ['No', 'Yes']
trans1 = ColumnTransformer([
    ('tr1',OneHotEncoder(categories=[l1],drop='first',dtype='int'),['education
    ('tr2',OneHotEncoder(categories=[l2],drop='first',dtype='int'),['self_empl
],remainder='passthrough')
trans1
```

Out[13]:

```
▸ ColumnTransformer
  ▸ tr1          ▸ tr2          ▸ remainder
    ▸ OneHotEncoder   ▸ OneHotEncoder   ▸ passthrough
```

In [14]:

```
# df.education.unique()
# df.self_employed.unique()
```

In [15]:

```
x = df.iloc[:,0:-1]
y = df.loan_status

# train test split
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat

from sklearn.ensemble import RandomForestClassifier
rdc = RandomForestClassifier(max_depth=3,n_jobs=-1,n_estimators=101)
rdc
```

Out[15]:

```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=3, n_estimators=101, n_jobs=-1)
```

In [16]:

```
# fit

x_train = pd.DataFrame(trans1.fit_transform(x_train),columns=trans1.get_featur
x_test = pd.DataFrame(trans1.transform(x_test),columns=trans1.get_feature_name
```

In [17]:

```
rdc.fit(x_train,y_train)
```

Out[17]:

```
▾ RandomForestClassifier
RandomForestClassifier(max_depth=3, n_estimators=101, n_jobs=-1)
```

In [18]:

```
y_pred = rdc.predict(x_test)

from sklearn.metrics import accuracy_score,precision_score,f1_score

print('accuracy',accuracy_score(y_test,y_pred))
print('precision',precision_score(y_test,y_pred))
print('f1_score',f1_score(y_test,y_pred))
```

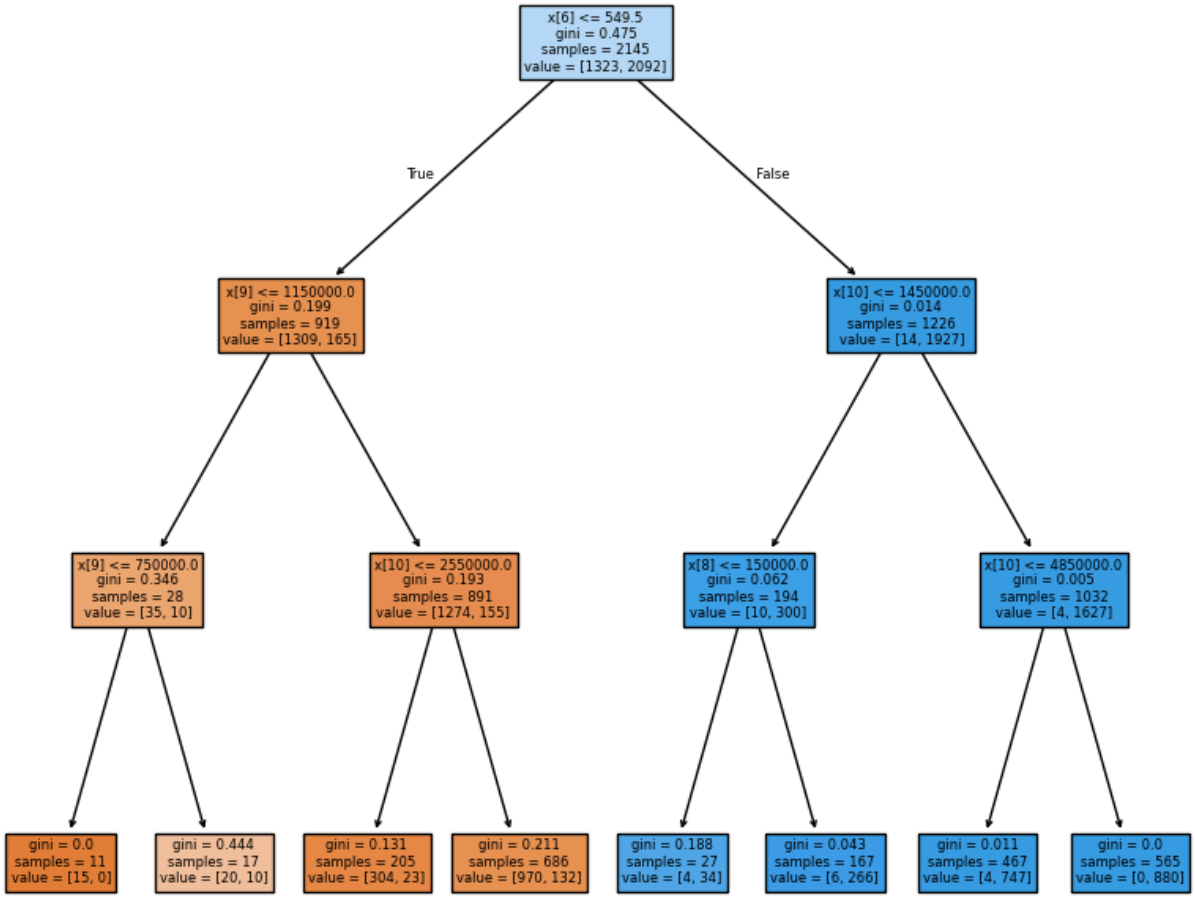
```
accuracy 0.9625292740046838
precision 0.9941176470588236
f1_score 0.9694072657743786
```

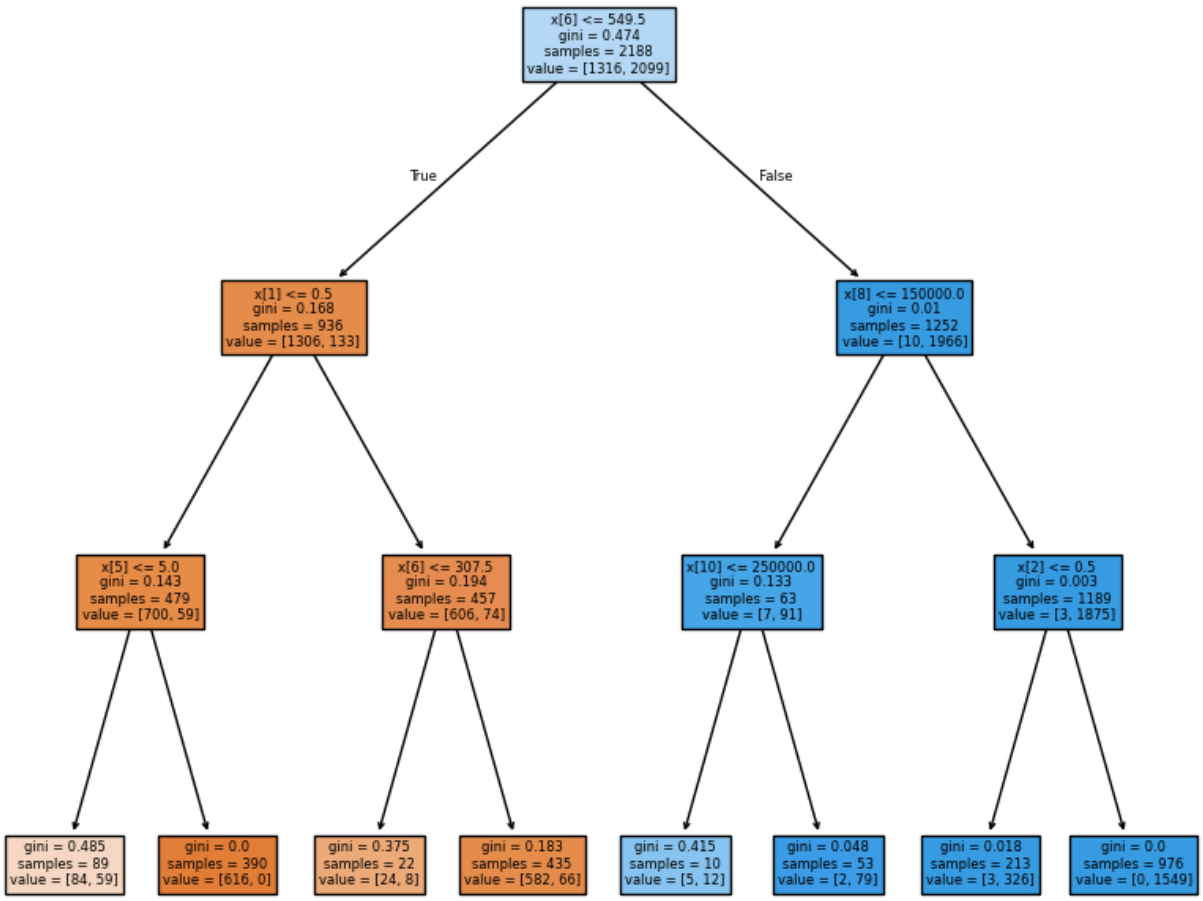
In [19]:
```
# fitting

rdc.score(x_train,y_train)
```

Out[19]: 0.9642752562225476

In [20]:
```
from sklearn import tree
for i in range(2):
    plt.figure(figsize=(10,9))
    tree.plot_tree(rdc[i],filled=True)
    plt.show()
    print('-'*120)
```





In [21]:
```
rdc.feature_importances_
```

Out[21]:
```
array([7.10728223e-05, 7.55272355e-05, 2.96279874e-03, 8.58064968e-03,
       1.99787542e-02, 1.42040157e-01, 7.70043182e-01, 1.83811748e-02,
       1.18411506e-02, 1.39951639e-02, 1.20303690e-02])
```

In [22]:
```
temp = pd.Series(rdc.feature_importances_,index=x_train.columns).reset_index()
```

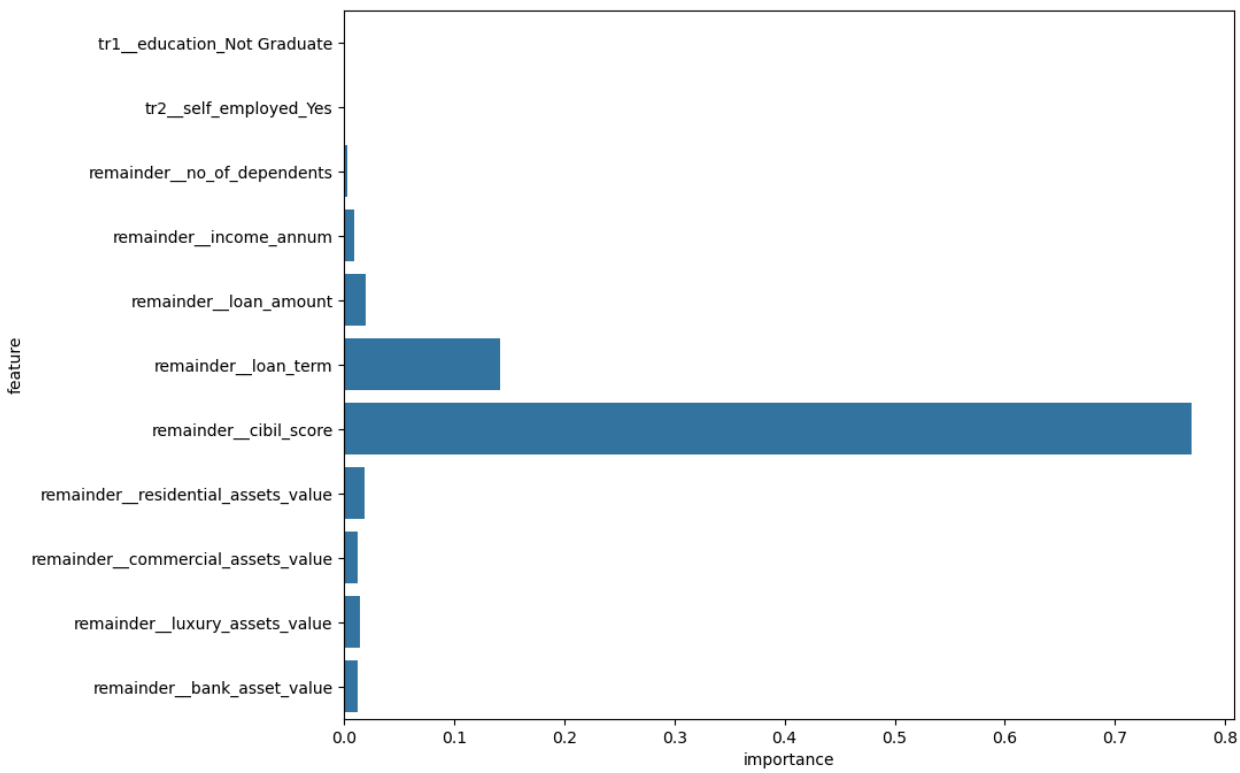
In [23]:
```
temp.columns = ['feature','importance']
temp
```

Out[23]:

	feature	importance
0	tr1__education_Not Graduate	0.000071
1	tr2__self_employed_Yes	0.000076
2	remainder__no_of_dependents	0.002963
3	remainder__income_annum	0.008581
4	remainder__loan_amount	0.019979
5	remainder__loan_term	0.142040
6	remainder__cibil_score	0.770043
7	remainder__residential_assets_value	0.018381
8	remainder__commercial_assets_value	0.011841
9	remainder__luxury_assets_value	0.013995
10	remainder__bank_asset_value	0.012030

In [24]:

```
plt.figure(figsize=(10,8))
sns.barplot(data=temp,x='importance',y='feature')
plt.show()
```



In [25]:

```
# classification

m1 = RandomForestClassifier(criterion='entropy',max_depth=4,
                            min_samples_split=11,
```

```
                                min_samples_leaf=10,
                                ccp_alpha=0.1)
m2 = RandomForestClassifier(criterion='entropy',
                                max_depth=10,min_samples_split=51,
                                min_samples_leaf=2,max_features='log2')

# criterion , gini, entropy  --> classification
# criterion, squared_error, absolute_error
# max_features --> sqrt
# None, sqrt, log2, 20
# 100 ---> 10
# log2 --> log2(100) --> 6.64 --> 6
# splitter
# ccp alpha --> pruning
# 0.01--0.1
```

In [26]:
```
m1.fit(x_train,y_train)
m2.fit(x_train,y_train)
```

Out[26]:
```
RandomForestClassifier
RandomForestClassifier(criterion='entropy', max_depth=10, max_feature
s='log2',
                       min_samples_leaf=2, min_samples_split=51)
```

In [27]:
```
print('bias',m1.score(x_train,y_train))
print('var',accuracy_score(y_test,m1.predict(x_test)))
print('-'*50)
print('bias',m2.score(x_train,y_train))
print('var',accuracy_score(y_test,m2.predict(x_test)))
```

```
bias 0.9537335285505124
var 0.9601873536299765
--------------------------------------------------
bias 0.9862371888726208
var 0.9718969555035128
```

In [28]:
```
# grid search cv
from sklearn.model_selection import GridSearchCV #seedha
#smartwork
from sklearn.model_selection import RandomizedSearchCV
```

In [29]:
```
param_dist = {
    'criterion':['gini','entropy'],
    'max_depth':[3,4,5,6,7,8,9,None],
    'ccp_alpha':[0,0.1,0.1],
    'min_samples_split':[10,20,30,40,50,60,100,None],
    'min_samples_leaf':[5,10,20,30,40,50,None]
}
```

In [33]:
```
m3 = RandomForestClassifier()
```

```
grid = GridSearchCV(m3,param_grid=param_dist,cv=5)
grid_ran=RandomizedSearchCV(estimator=m3,param_distributions=param_dist,n_iter
grid_ran
```

Out[33]:

```
▸ RandomizedSearchCV
▸ estimator: RandomForestClassifier
  ▸ RandomForestClassifier
```

In [34]:

```
#grid.fit(x_train,y_train)
grid_ran.fit(x_train,y_train)
```

Out[34]:

```
▸ RandomizedSearchCV
▸ best_estimator_: RandomForestClassifier
  ▸ RandomForestClassifier
```

In [35]:

```
grid_ran.best_estimator_
grid_ran.best_score_
grid_ran.best_params_
```

Out[35]:

```
{'min_samples_split': 60,
 'min_samples_leaf': 50,
 'max_depth': None,
 'criterion': 'entropy',
 'ccp_alpha': 0.1}
```

In [36]:

```
# pipelines

from sklearn.pipeline import Pipeline

trans = ColumnTransformer([
    ('tr1',OneHotEncoder(categories=[l1],drop='first',dtype='int'),['education
    ('tr2',OneHotEncoder(categories=[l2],drop='first',dtype='int'),['self_empl
],remainder='passthrough')
trans
```

Out[36]:

```
▸ ColumnTransformer
  ▸ tr1          ▸ tr2          ▸ remainder
    ▸ OneHotEncoder   ▸ OneHotEncoder   ▸ passthrough
```

In [37]:

```
estimator = RandomForestClassifier(min_samples_split=30,
 min_samples_leaf=50,
 max_depth=None,
 criterion='entropy',
 ccp_alpha=0.1)
estimator
```

Out[37]:

```
▾ RandomForestClassifier
RandomForestClassifier(ccp_alpha=0.1, criterion='entropy', min_sample
s_leaf=50,
                       min_samples_split=30)
```

In [39]:

```
Pipeline([
    ('transform',trans),
    ('estimator',estimator)
])
```

Out[39]:

```
▸ Pipeline
  ▸ transform: ColumnTransformer
    ▸ tr1          ▸ tr2          ▸ remainder
      ▸ OneHotEncoder   ▸ OneHotEncoder   ▸ passthrough
  ▸ RandomForestClassifier
```

In [ ]: